

Technical Document #551: Natural Language Processing - Comprehensive Overview

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Word embeddings represent words as dense vectors in continuous space. Word2Vec, GloVe, and FastText are popular word embedding techniques.

Machine translation converts text from one language to another. Neural machine translation using encoder-decoder architectures has achieved human-level performance.

Tokenization is the process of breaking text into individual units called tokens. Subword tokenization methods like BPE handle out-of-vocabulary words gracefully.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Machine translation converts text from one language to another.
- Language models predict the probability of a sequence of words.
- Named entity recognition (NER) identifies and classifies named entities in text into predefined categories such as person names.